

Q#1. Evaluate

$$\begin{aligned} \text{(i)} \quad i^{10} \cdot (-i^{12}) &= (i^2)^5 \cdot (-i^2)^6 \\ &= (-1)^5 \cdot (-(-1)^6) \\ &= (-1)(-1) = 1 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad (-i)^{16} \cdot (-i)^6 &= (-1)^{16} (i^2)^8 (-1)^6 (i^2)^3 \\ &= (1)(-1)^8 (1)(i^2)^3 \\ &= (1)(-1)^3 = -1 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad (i)^{11} \cdot (-i)^{14} &= i^{10} \cdot (-1)^4 (i)^1 \\ &= i(i^2)^5 \cdot (1)(i^2)^7 \\ &= i(-1)^5 (-1)^7 \\ &= (-i)(-1) = i \end{aligned}$$

Q#2. For what value of n , i^n is equal to 1, i , -1 , or $-i$ where $1 \leq n \leq 4$

$$i^n = 1$$

$$i^4 = 1$$

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = i^2 \cdot i = -i$$

Q#3. Simplify the following.

$$\text{(i)} \quad (-6, 3) \cdot (4, -2)$$

$$\begin{aligned} &= (-6 + 3i) \cdot (4 - 2i) \\ &= -6(4 - 2i) + 3i(4 - 2i) \\ &= -24 + 12i + 12i - 6i^2 \end{aligned}$$

Exercise 1.1

$$\begin{aligned} &= -24 + 24i + 6 \\ &= -18 + 24i \end{aligned}$$

Method II

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc)$$

$$\begin{aligned} (-6, 3) \cdot (4, -2) &= (-6)(4) - (3)(-2), (-6)(-2) + (3)(4) \\ &= (-24 + 6, 12 + 12) \\ &= (-18, 24) \end{aligned}$$

$$\text{(ii)} \quad (8, -4) \div (-2, 2)$$

Method 1

$$\begin{aligned} &= \frac{8 - 4i}{-2 + 2i} \\ &= \frac{8 - 4i}{-2 + 2i} \times \frac{-2 - 2i}{-2 - 2i} \\ &= \frac{8(-2 - 2i) - 4i(-2 - 2i)}{(-2)^2 - (2i)^2} \end{aligned}$$

$$\begin{aligned} &= \frac{-16 - 8i - 8}{4 - 4i^2} \\ &= \frac{-16 - 8i - 8}{4 + 4} \end{aligned}$$

$$\begin{aligned} &= \frac{-24 - 8i}{4 + 4} \\ &= \frac{-24 - 8i}{8} \end{aligned}$$

$$\begin{aligned} &= \frac{-24}{8} - \frac{8i}{8} \\ &= -3 - i \end{aligned}$$

$$\begin{aligned} &= -3 - i \end{aligned}$$

Method II

$$(8, -4) \div (-2, 2)$$

$$\begin{aligned} (a, b) \div (c, d) &= \left(\frac{ac + bd}{c^2 + d^2}, \frac{bc - ad}{c^2 + d^2} \right) \end{aligned}$$

$$\begin{aligned} &= \left(\frac{(8)(-2) + (-4)(2)}{(-2)^2 + (2)^2}, \frac{(-4)(-2) - (8)(2)}{(-2)^2 + (2)^2} \right) \end{aligned}$$

$$\begin{aligned} &= \left(\frac{-16 - 8}{8}, \frac{+8 - 16}{8} \right) = (-3, -1) \end{aligned}$$

$$\text{(iii)} \quad (2, 3) \div (4, 5)$$

$$\begin{aligned} &= \frac{2 + 3i}{4 + 5i} \end{aligned}$$

$$\begin{aligned} &= \frac{2 + 3i}{4 + 5i} \times \frac{4 - 5i}{4 - 5i} \end{aligned}$$

$$\begin{aligned} &= \frac{8 - 10i + 12i - 15i^2}{16 - 25i^2} \end{aligned}$$

$$\begin{aligned} &= \frac{23 + 2i}{41} \end{aligned}$$

$$\begin{aligned} &= \left(\frac{23}{41}, \frac{2}{41} \right) \end{aligned}$$

$$\text{(iv)} \quad (6 + 5i) - (4 + 3i)$$

$$\begin{aligned} &= 2 + 2i \end{aligned}$$

$$\begin{aligned} \text{(v)} \quad (5 - 6i) + (3 + 4i) - (5i - 7) &= (5 - 6i) + (3 + 4i) - (-7 + 5i) \\ &= 15 - 7i \end{aligned}$$

$$\text{(vi)} \quad (4 + 5i)(6 + 7i)$$

$$\begin{aligned} &= 24 + 28i + 30i + 35i^2 \\ &= 24 - 35 + 58i \\ &= -11 + 58i \end{aligned}$$

$$\text{(vii)} \quad (7 + 4i) \div (8 + 5i)$$

$$\begin{aligned} &= \frac{7 + 4i}{8 + 5i} \end{aligned}$$

$$\begin{aligned} &= \frac{7 + 4i}{8 + 5i} \times \frac{8 - 5i}{8 - 5i} \end{aligned}$$

$$\begin{aligned} &= \frac{56 - 35i + 32i - 20i^2}{8^2 - (5i)^2} \end{aligned}$$

$$\begin{aligned} &= \frac{76 - 3i}{64 + 25} = \frac{76}{89} - \frac{3}{89}i \end{aligned}$$

viii) $(5, -6) + (4, 8) - (3, -2)$
 $= (5+4-3, -6+8+2)$
 $= (6, 4)$

5) Simplify

i) $(2-i)^4$
 $= [(2-i)^2]^2$
 $= [4-4i+i^2]^2$
 $= [4-4i-1]^2$
 $= [3-4i]^2$
 $= 9-24i+16i^2$
 $= 9-16-24i$
 $= -7-24i$

ii) $(-1-i\sqrt{3})^2$
 $= (-1)^2 - 2(-1)(i\sqrt{3}) + (i\sqrt{3})^2$
 $= 1 + 2\sqrt{3}i + 3i^2$
 $= -2 + 2\sqrt{3}i$

iii) $(-1+i\sqrt{3})^2$
 $= (-1)^2 + 2(-1)(i\sqrt{3}) + (i\sqrt{3})^2$
 $= 1 - 2\sqrt{3}i + 3i^2$
 $= -2 - 2\sqrt{3}i$

iv) $(1+i)^3$
 $= 1^3 + 3(1)^2i + 3(1)i^2 + i^3$
 $= 1 + 3i - 3 + i^2$
 $= -2 + 3i - 1$
 $= -3 + 3i$

v) $\frac{\sqrt{2}+i}{\sqrt{2}-i}$

$$= \frac{\sqrt{2}+i}{\sqrt{2}-i} \times \frac{\sqrt{2}+i}{\sqrt{2}+i}$$

$$= \frac{(\sqrt{2}+i)^2}{(\sqrt{2})^2 - (i)^2} = \frac{2+2\sqrt{2}i+i^2}{2-(-1)}$$

$$= \frac{1+2\sqrt{2}i}{3} = \frac{1}{3} + \frac{2\sqrt{2}i}{3}$$

vi) $\frac{1+i}{1-i} \cdot \frac{2-i}{1-i}$
 $= \frac{(1+i)(2-i)}{(1-i)^2}$
 $= \frac{2-i+2i-i^2}{1-2i+i^2} = \frac{3+i}{-2i}$
 $= \frac{3+i}{-2i} \times \frac{0+2i}{0+2i} = \frac{6i+2i^2}{-4i^2}$
 $= \frac{-2+6i}{4} = -\frac{1}{2} + \frac{3}{2}i$

vii) $\frac{(2+i)^2}{3-4i}$
 $= \frac{4+4i+i^2}{3-4i} = \frac{3+4i}{3-4i}$
 $= \frac{3+4i}{3-4i} \times \frac{3+4i}{3+4i} = \frac{(3+4i)^2}{9-(4i)^2}$
 $= \frac{9+24i+16i^2}{9-16i^2}$
 $= \frac{9-16+24i}{9+16} = \frac{-7+24i}{25}$
 $= -\frac{7}{25} + \frac{24}{25}i$

viii) $\frac{1}{(2-i)^2}$
 $= \frac{1}{4-4i+i^2} = \frac{1}{3-4i}$
 $= \frac{1}{3-4i} \times \frac{3+4i}{3+4i}$

$$= \frac{3+4i}{3^2 - (4i)^2} = \frac{3+4i}{9+16}$$

$$= \frac{3}{25} + \frac{4}{25}i$$

Q#5 Show That $z=1+i$
Satisfies The equation
 $z^2 - 2z + 2 = 0$

For $z=1+i$

$$\Rightarrow (1+i)^2 - 2(1+i) + 2 = 0$$

$$\Rightarrow 1+2i+i^2 - 2 - 2i + 2 = 0$$

$$\Rightarrow \cancel{1} + \cancel{2i} - \cancel{1} - \cancel{2} - \cancel{2i} + \cancel{2} = 0$$

$$0 = 0$$

For $z=1-i$

$$(1-i)^2 - 2(1-i) + 2 = 0$$

$$\Rightarrow 1^2 - 2i + i^2 - 2 + 2i + 2 = 0$$

$$\Rightarrow \cancel{1} - \cancel{2i} - \cancel{1} - \cancel{2} + \cancel{2i} + \cancel{2} = 0$$

$$0 = 0$$

Q#6. Find The Conjugate
and absolute value of
The following

i) $z = 4+5i$

$$\bar{z} = 4-5i$$

$$|z| = \sqrt{(4)^2 + (5)^2}$$

$$|z| = \sqrt{16+25} = \sqrt{41}$$

ii) $z = -1+7i$

$$\bar{z} = -1-7i$$

$$|z| = \sqrt{(-1)^2 + (7)^2} = \sqrt{50}$$

$$|z| = 5\sqrt{2}$$

iii) $z = 0 + \sqrt{3}i$ Pg 3

$$\bar{z} = 0 - \sqrt{3}i = -\sqrt{3}i$$

$$|z| = \sqrt{0^2 + (\sqrt{3})^2} = 3$$

iv) $\sqrt{7}-3i$

$$\bar{z} = \sqrt{7}+3i$$

$$|z| = \sqrt{(\sqrt{7})^2 + (3)^2}$$

$$|z| = \sqrt{7+9} = \sqrt{16} = +4$$

v) $-3-4i$

$$z = -3-4i$$

$$\bar{z} = -3+4i$$

$$|z| = \sqrt{(-3)^2 + (-4)^2} = \sqrt{9+16} = 5$$

vi) $z = (5-4i)^2$

$$z = 25 - 40i + 16i^2$$

$$z = 9 - 40i$$

$$\bar{z} = 9 + 40i$$

$$|z| = \sqrt{9^2 + 40^2} = \sqrt{81+1600}$$

$$|z| = 41$$

vii) $z = \frac{2}{3} - \sqrt{\frac{-9}{16}} = \frac{2}{3} - \frac{3}{4}i$

$$\bar{z} = \frac{2}{3} + \frac{3}{4}i$$

$$|z| = \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{3}{4}\right)^2} = \sqrt{\frac{4}{9} + \frac{9}{16}}$$

$$|z| = \sqrt{\frac{16 \times 4 + 9 \times 9}{9 \times 16}} = \sqrt{\frac{145}{144}}$$

$$|z| = \frac{\sqrt{145}}{12}$$

viii) $z = \frac{(1+i)(1+2i)}{3+i}$

$$z = \frac{1+2i+i+2i^2}{3+i}$$

$$z = \frac{-1+3i}{3+i}$$

$$z = \frac{-1+3i}{3+i} \times \frac{3-i}{3-i}$$

$$z = \frac{-3+i+9i-3i^2}{9-i^2}$$

$$z = \frac{10i}{10} = i$$

$$\bar{z} = -i$$

$$|z| = \sqrt{0^2 + 1^2} = 1$$

Q#7 Find real and Imaginary Parts of

$$\textcircled{i} z = 2i(3-5i)$$

$$z = 6i - 10i^2$$

$$z = 10 + 6i$$

$$\text{Re}(z) = 10$$

$$\text{Im}(z) = 6$$

$$\textcircled{ii} z = \frac{\sqrt{5}+i}{\sqrt{5}-i}$$

$$z = \frac{\sqrt{5}+i}{\sqrt{5}-i} \times \frac{\sqrt{5}+i}{\sqrt{5}+i}$$

$$z = \frac{(\sqrt{5}+i)^2}{(\sqrt{5})^2 - i^2}$$

$$z = \frac{5+2\sqrt{5}i+i^2}{5+1}$$

$$z = \frac{4+2\sqrt{5}i}{6}$$

$$z = \frac{2}{3} + \frac{\sqrt{5}}{3}i$$

$$\text{Re}(z) = \frac{2}{3}$$

$$\text{Im}(z) = \frac{\sqrt{5}}{3}$$

Q#8 if $z = x+iy$ (Pg 4)

where $\text{Re}(z) = 0$ and $|z| = 2$
Then find z

$$\text{Re}(z) = \text{Re}(x+iy) = 0$$

$$x = 0$$

$$|z| = \sqrt{x^2 + y^2} = 2$$

$$\Rightarrow |z| = \sqrt{0^2 + y^2} = 2$$

$$\Rightarrow |z| = y = \pm 2$$

$$z = 0 \pm 2i$$

Q#9 Solve the following Complex equations.

$$\textcircled{i} (x+iy)(2+3i) = (-4+7i)$$

$$\Rightarrow (x+iy)(2+3i) = -4+7i$$

$$\Rightarrow 2x+3xi+2yi+3yi^2 = -4+7i$$

$$\Rightarrow (2x-3y) + (3x+2y)i = -4+7i$$

$$2x-3y = -4 \quad \text{--- (1)}$$

$$3x+2y = 7 \quad \text{--- (2)}$$

$$\text{eq (1)} \times 2 \quad 4x-6y = -8$$

$$\text{eq (2)} \times 3 \quad 9x+6y = 21 \quad +$$

$$13x = 13$$

$$\boxed{x=1}$$

$$\textcircled{i} \Rightarrow 2(1) - 3y = -4$$

$$-3y = -4-2 = -6$$

$$\boxed{y=2}$$

$$\textcircled{ii} (x+3i)^2 = 2yi$$

$$\Rightarrow x^2 + 6xi + 9i^2 = 2yi$$

$$\Rightarrow x^2 - 9 + 6xi = 2yi$$

$$x^2 - 9 = 0, \quad 6x = 2y$$

$$x^2 = 9 \quad 3x = y$$

$$x = \pm 3$$

$$y = +9, y = -9$$

Q#10. Represent the following complex numbers on Complex plane

① $(2, -3)$



② $(3, 4)$



③ $(-5, 7)$

$(-5, 7)$



④ $(-6, -2)$



⑤ $(0, 6)$

$(0, 6)$



⑥ $(-5, 0)$



Q#11 Find the length of vector representing the complex numbers.

① $-5 + 2i$

$$|Z| = \sqrt{(-5)^2 + (2)^2} = \sqrt{25 + 4}$$

$$|Z| = \sqrt{29}$$

② $Z = \frac{7}{3} + \frac{8}{3}i$

$$|Z| = \sqrt{\left(\frac{7}{3}\right)^2 + \left(\frac{8}{3}\right)^2}$$

$$|Z| = \sqrt{\frac{49}{9} + \frac{64}{9}} = \sqrt{\frac{49+64}{9}}$$

$$|Z| = \frac{\sqrt{113}}{3}$$

③ $Z = \frac{1+i}{\sqrt{2}}$

$$|Z| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2}$$

$$|Z| = \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{1} = 1$$

④

$$Z = \frac{1+\sqrt{3}i}{2}$$

$$|Z| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$|Z| = \sqrt{\frac{1}{4} + \frac{3}{4}} = \sqrt{1} = 1$$